

Course (TY03020): (VETERINARY CLINICAL PHARMACOLOGY)

1. General information

- Term: 09
- Credits: **Total credits 02 (Lecture: 1.5 – Practice: 0.5)**
- **Self-study: 06** credits
- Credit hours for teaching and learning activities: 1500 hrs
- Self-study: 4500 hrs.
- Department conducting the course:
 - Department: Pharmacology, Internal Medicine, and Diagnostic
 - Faculty: Veterinary Medicine
- Kind of the course:

Foundation ☐		Fundamental ☒		Option 1 ☐	
Compulsory	Elective	Compulsory	Elective	Compulsory	Elective
☐	☐	☐	☒	☐	☐

- Prerequisite course(s): TY02006_Veterinary Pharmacology

2. Course objectives and expected learning outcomes

* *Course objectives:*

- Knowledge: Course provided for students with knowledge on how to select, use and manage drugs in veterinary clinical practice to promote desired effects and limit risks to animals, humans, and the environment and optimize economic efficiency.
- Skills: Course provide students with skills to analyze and handle situations, to make plan, to make decisions about medical using in veterinary clinical practice.
- Attitude: Course provide students with attitudes to be proactive, confident, flexible, and respectful of animals and the environment when using medicines in veterinary clinical practice.

* *Course expected learning outcomes*

Notation	Course expected learning outcomes (CLOs) After successfully completing this course, students are able to	Program performance criteria (PPC)
Knowledge		
CLO1	Applying pharmacokinetic and pharmacodynamic parameters to optimize dosage regimens in veterinary medicine	3.2
CLO2	Detecting adverse reactions, risk of drug resistance, possibility of residues of veterinary drugs for a treatment regimen for animals	5.4
Skills		
CLO3	Clearly indicate the causes, duration, and pathogenesis of various clinical states in veterinary medicines	5.4, 8.2
CLO4	Guidelines for the selection and combination of drug to	9.1, 9.3

	develop a drug therapy based on the clinical situation in veterinary medicines	
Attitude		
CLO5	Demonstrate the ability to adapt and synthesize according to the regulations of the subject. Organize and connect learning team members to build teamwork with respect for partners with independent thinking. Comply with the provisions of veterinary laws and related regulations	12.1

3. Course description

TY03020. Veterinary Clinical Pharmacology. (2 Credits: 1-0,5-06). This course consists of introduction to the basic principles that help in the selection, use and management of drugs in veterinary clinical practice, including applied pharmacokinetics, pharmacokinetic parameters, adverse effects, veterinary drug information, antibiotic resistance control, drug interactions and principles of building a drug therapy for prevention and treatment in veterinary medicine.

Teaching method: Students listen to lectures in class in combination with self-study, self-referencing documents and exchanging with friends and teachers. Assessment method: 10% attendance, 20% practical exam, 20% presentation and assignment, 50% final exam.

4. Teaching and learning & assessment methods

CLOs	CLO1	CLO2	CLO3	CLO4	CLO5
Teaching methods					
Lecturing	x				
Teaching through practical work			x	x	x
Case study		x	x	x	
Assessment					
Rubric 1. Attendance (10 %)					x
Rubric 2. Practical (20 %)			x	x	x
Rubric 3. Presentation and assignment (20 %)					
Rubric 4. Final exam (50 %)		x	x	x	

5. Student tasks

-Attendance: All students will be expected to attend at least 70% of lecturing.

-Preparation for the lecture: All students take notes on these content outlines before each class

- Practices: All students must fully participate of the subject practice sessions and ensure having personal protective equipment for subject practice

- Assignment: All students must engage in generating ideas, searching and processing information, preparing presentations, presenting and answering group discussion questions in class.

- Final exam: All students must fully participate in the subject practice sessions, group presentations, midterm exams and final exams.

6. Textbooks and references

* *Textbooks/Lecture Notes:*

Pham Khac Hieu (2019) *Molecular pharmacology and applications in veterinary medicine*, Vietnam National University of Agriculture Publishing.

* *Additional references:*

Jim E Riviere and Mark G Papich (2018) *Veterinary pharmacology and therapeutics*, 10th edition, Wily Blackwell Publishing.

7. Course outline

Week	Content	Course expected learning outcomes
1	Chapter 1: Basic principles of drug selection and drug using in veterinary clinical practice	
	A/ Main contents: (4 hours) 1. Theories: (4 hours) 1.1. Applied pharmacokinetics 1.2. Pharmacokinetic parameters and applications in veterinary clinical 1.3. Applied pharmacodynamics 1.4. Adverse drug reactions 2. Practice: (2 hours) Factors affecting the pharmacological effects of drugs	CLO1, 5
	B/ Self-study contents: (18 hours) 1.1. Read in advance the next lesson's lecture and the exercise 1.2. Prepare seminar (if in the presentation group) or learn about the part to be presented to participate in discussion and ask questions	
2	Chapter 2: Veterinary Medicines Information	
	A/ Main contents: (5 hours) 1. Theories: (3 hours) 2.1. The role of veterinary drug information 2.2. Veterinary medicine information source 2. Practice: (2 hours) Practice selection, arrangement, and management of medicine cabinets at commune, district, and retail levels Practice building and arranging medicine cabinets for pig and poultry farms	CLO1, 2, 5
	B/ Self-study contents: (15 hours) 2.1. Read in advance the next lesson's lecture and the exercise 2.2. Prepare seminar (if in the presentation group) or learn about the part to be presented to participate in discussion and ask questions	

3	Chapter 3: Use of antibiotics drugs in veterinary medicines	
	A/ Main contents: 1. Theories: (3 hours) 3.1. Antibiotic resistance of pathogenic bacteria 3.1.1. Causes 3.1.2. Bacterial resistance mechanism of antibiotic 3.1.3. Transfer of antibiotic resistance 3.1.4. Strategies to limit bacterial resistance of antibiotic in veterinary medicine 3.2. Application of antibiotics in veterinary medicines 3.2.1. Use of antibiotics in the treatment of respiratory tract infections 3.2.2. Use of antibiotics in the treatment of gastrointestinal infections 3.2.3. Use of antibiotics in the treatment of urogenital infections 3.2.4. Use of antibiotics for young and pregnant cattle 2. Practice: (2 hours) Practice choosing drugs to treat bacterial infections in veterinary medicine - Practice selecting drugs and developing treatment regimens for respiratory tract infections - Practice selecting drugs and developing treatment regimens for gastrointestinal infections Practice choosing drugs and developing treatment regimens for urogenital tract infections	CLO1, 2, 5
	B/ Self-study contents: (15 hours) Read in advance the next lesson's lecture and the exercise Prepare seminar (if in the presentation group) or learn about the part to be presented to participate in discussion and ask questions	
4	Chapter 4: Application of drugs to treat blood parasites and coccidiosis in veterinary medicines	
	A/ Main contents: 1. Theories: (6 hours) 4.1. Using drugs to treat blood sugar parasites in veterinary medicines 4.2. Use of coccidiosis drugs in veterinary medicines 2. Practice: (2 hours) Practice choosing drugs to treat veterinary parasites - Practice selecting drugs and developing treatment regimens for veterinary external parasites	CLO3, 5

	- Practice selecting drugs and developing treatment regimens for roundworms in animals - Practice selecting drugs and developing treatment regimens for tapeworms in veterinary medicine	
	B/ Self-study contents: (24 hours) 2.3. Read in advance the next lesson's lecture and the exercise Prepare seminar (if in the presentation group) or learn about the part to be presented to participate in discussion and ask questions	
	Chapter 5: Use of growth stimulants substances in animals	
5	A/ Main contents: (5 hours) 1. Theories: (3 hours) 5.1. Outline 5.2. Application of growth stimulants substances in animals 2. Practice: (2 hours) Practice selecting growth stimulants substances in animals	CLO3, 5
	B/ Self-study contents: (15 hours) Read in advance the next lesson's lecture and the exercise Prepare seminar (if in the presentation group) or learn about the part to be presented to participate in discussion and ask questions	